



AROR UNIVERSITY OF ART, ARCHITECTURE, DESIGN & HERITAGE, SUKKUR, SINDH

Course: Applied Physics

Lecture 02

Instructor:

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Product of Vectors

1. SCALAR PRODUCT
2. VECTOR PRODUCT

Product of Vectors

- We saw how vector addition develops naturally from the problem of combining displacements.
- It will prove useful for calculations with many other vector quantities.
- We can also express many physical relationships by using products of vectors.
- Vectors are not ordinary numbers, so we can't directly apply ordinary multiplication to vectors.
- We'll define two different kinds of products of vectors.
 1. The first, called the **scalar product**, yields a result that is a **scalar quantity**.
 2. The second, the **vector product**, yields another **vector**.

1. Scalar Product / Dot product

Scalar product of two vectors is defined as :

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

➤ It is actually product of two components:

1. Magnitude of 1st vector $\vec{A}$
2. Magnitude of component 2nd vector $\vec{B}$ in the direction of 1st vector

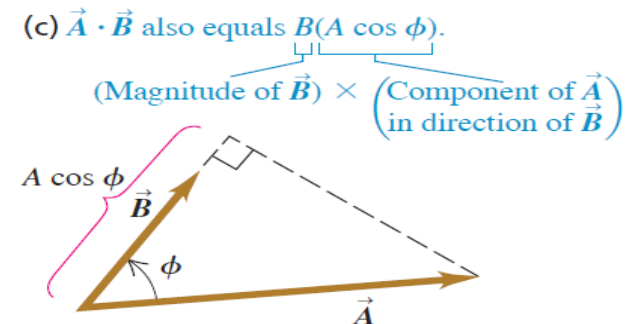
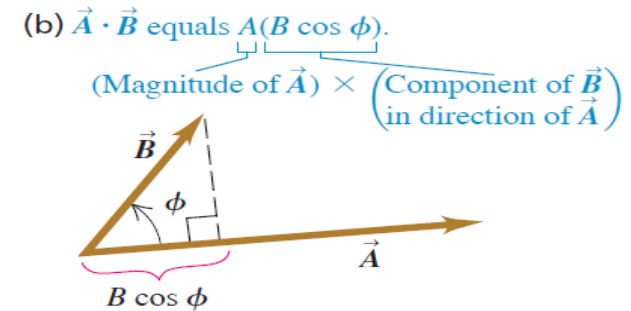
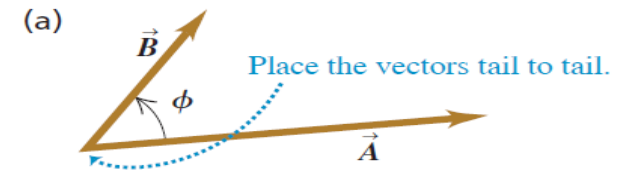
$$AB_x$$

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

➤ Example:

$$W = \vec{F} \cdot \vec{d}$$

$$P = \vec{F} \cdot \vec{v}$$

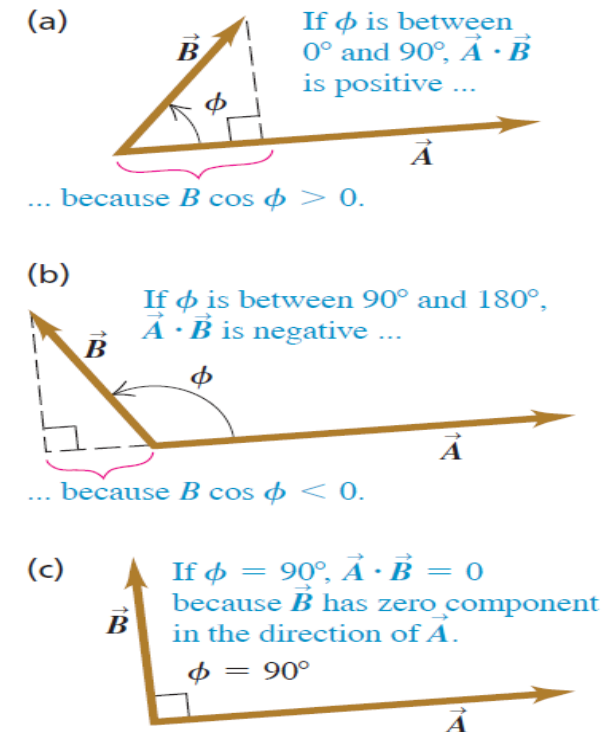


Scalar Product / Dot product (2)

A. Characteristics of scalar product:

- i. The scalar product is a scalar quantity, not a vector, and it may be positive, negative, or zero.
- ii. When θ is between 0 and 90, $\cos\theta > 0$ and the scalar product is positive.
- iii. When θ is between 90 and 180 so $\cos\theta < 0$, the component of B in the direction of A is negative, and $\vec{A} \cdot \vec{B}$ is negative.
- iv. Finally, when $\theta = 90$ then $\vec{A} \cdot \vec{B} = 0$
- v. The scalar product of two perpendicular vectors is always zero.

Figure 1.27 The scalar product $\vec{A} \cdot \vec{B} = AB \cos \phi$ can be positive, negative, or zero, depending on the angle between $\vec{A}$ and $\vec{B}$.



Scalar Product / Dot product (3)

B. Using Components to Calculate the Scalar Product

- We can calculate the scalar product $\vec{A} \cdot \vec{B}$ directly if we know the x-, y-, and z-component.
- To see how this is done, let's first work out the scalar products of the unit vectors $\hat{i}$, $\hat{j}$ and $\hat{k}$.
- All unit vectors have magnitude 1 and are perpendicular to each other
$$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = (1)(1) \cos 0^\circ = 1$$
$$\hat{i} \cdot \hat{j} = \hat{i} \cdot \hat{k} = \hat{j} \cdot \hat{k} = (1)(1) \cos 90^\circ = 0$$
- Now we express $\vec{A}$ and $\vec{B}$ in terms of their components, expand the product, and use these products of unit vectors:

$$\begin{aligned}\vec{A} \cdot \vec{B} &= (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) \cdot (B_x \hat{i} + B_y \hat{j} + B_z \hat{k}) \\&= A_x \hat{i} \cdot B_x \hat{i} + A_x \hat{i} \cdot B_y \hat{j} + A_x \hat{i} \cdot B_z \hat{k} \\&\quad + A_y \hat{j} \cdot B_x \hat{i} + A_y \hat{j} \cdot B_y \hat{j} + A_y \hat{j} \cdot B_z \hat{k} \\&\quad + A_z \hat{k} \cdot B_x \hat{i} + A_z \hat{k} \cdot B_y \hat{j} + A_z \hat{k} \cdot B_z \hat{k} \\&= A_x B_x \hat{i} \cdot \hat{i} + A_x B_y \hat{i} \cdot \hat{j} + A_x B_z \hat{i} \cdot \hat{k} \\&\quad + A_y B_x \hat{j} \cdot \hat{i} + A_y B_y \hat{j} \cdot \hat{j} + A_y B_z \hat{j} \cdot \hat{k} \\&\quad + A_z B_x \hat{k} \cdot \hat{i} + A_z B_y \hat{k} \cdot \hat{j} + A_z B_z \hat{k} \cdot \hat{k}\end{aligned}$$

Scalar Product / Dot product (4)

- From Eqs. (1.17) you can see that six of these nine terms are zero. The three that survive
Give

Scalar (dot) product
of vectors $\vec{A}$ and $\vec{B}$

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

Components of $\vec{A}$

Components of $\vec{B}$

The diagram illustrates the scalar product of two vectors $\vec{A}$ and $\vec{B}$. It shows the equation $\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$. Above the equation, a bracket groups the vectors $\vec{A}$ and $\vec{B}$ with the text "Scalar (dot) product of vectors $\vec{A}$ and $\vec{B}$ ". Below the equation, two sets of dashed blue arrows point from the components of $\vec{A}$ (A_x, A_y, A_z) to the corresponding components of $\vec{B}$ (B_x, B_y, B_z). The top set of arrows is labeled "Components of $\vec{A}$ " and the bottom set is labeled "Components of $\vec{B}$ ".

- Thus the scalar product of two vectors is the sum of the products of their respective components.
- The scalar product gives a straightforward way to find the angle θ between any two vectors $\vec{A}$ and $\vec{B}$ whose components are known. In this case we can use above Eq. to find the scalar product of $\vec{A}$ and $\vec{B}$.

Scalar Product / Dot product (5)

Numerical: Find the angle between the vectors in terms of scalar product

Find the angle between the vectors

$$\vec{A} = 2.00\hat{i} + 3.00\hat{j} + 1.00\hat{k}$$

and

$$\vec{B} = -4.00\hat{i} + 2.00\hat{j} - 1.00\hat{k}$$

Solution

$$\cos \phi = \frac{\vec{A} \cdot \vec{B}}{AB} = \frac{A_x B_x + A_y B_y + A_z B_z}{AB}$$

We can use this formula to find the angle between *any* two vectors $\vec{A}$ and $\vec{B}$. Here we have $A_x = 2.00$, $A_y = 3.00$, and $A_z = 1.00$, and $B_x = -4.00$, $B_y = 2.00$, and $B_z = -1.00$. Thus

→ →

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

$$= (2.00)(-4.00) + (3.00)(2.00) + (1.00)(-1.00)$$

$$= -3.00$$

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2} = \sqrt{(2.00)^2 + (3.00)^2 + (1.00)^2}$$

$$= \sqrt{14.00}$$

$$B = \sqrt{B_x^2 + B_y^2 + B_z^2} = \sqrt{(-4.00)^2 + (2.00)^2 + (-1.00)^2}$$

$$= \sqrt{21.00}$$

$$\cos \phi = \frac{A_x B_x + A_y B_y + A_z B_z}{AB} = \frac{-3.00}{\sqrt{14.00} \sqrt{21.00}} = -0.175$$

$$\phi = 100^\circ$$

2. Vector Product/ Cross Product:

Vector product of two vectors is defined as;

$$\vec{A} \times \vec{B} = AB \sin \theta$$

➤ It is actually product of two components:

1. Magnitude of 1st vector $\vec{A}$
2. Magnitude of component 2nd vector $\vec{B}$ perpendicular to of 1st vector

$$AB_y$$

$$\vec{A} \times \vec{B} = AB \sin \theta$$

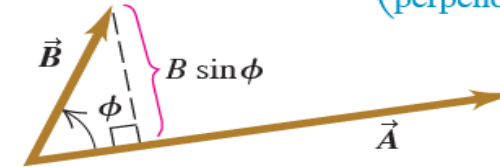
➤ The result of cross product is also vector quantity, because the cross product is the product of magnitude of 1st vector with the Magnitude of component 2nd vector $\vec{B}$ perpendicular to of 1st vector, therefore we need an other direction to get cross product.

Figure 1.31 Calculating the magnitude $AB \sin \phi$ of the vector product of two vectors, $\vec{A} \times \vec{B}$.

(a)

(Magnitude of $\vec{A} \times \vec{B}$) equals $A(B \sin \phi)$.

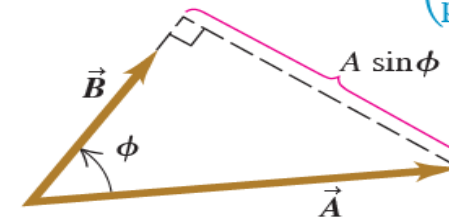
(Magnitude of $\vec{A}$) $\times$ (Component of $\vec{B}$ perpendicular to $\vec{A}$)



(b)

(Magnitude of $\vec{A} \times \vec{B}$) also equals $B(A \sin \phi)$.

(Magnitude of $\vec{B}$) $\times$ (Component of $\vec{A}$ perpendicular to $\vec{B}$)



Vector Product/ Cross Product (2)

A. Characteristics of vector product:

- i. We measure the angle θ from $\vec{A}$ toward $\vec{B}$ and take it to be the smaller of the two possible angles, so θ ranges from 0 to 180 degree. Then $\sin\theta \geq 0$ and $AB\sin\theta$ is never negative, as must be the case for a vector magnitude.
- ii. Note that when $\vec{A}$ and $\vec{B}$ are parallel or antiparallel, $\theta = 0^\circ$ or 180° and $AB\sin\theta = 0$.
 - That is, *the vector product of two parallel or antiparallel vectors is always zero.*
 - In particular, *the vector product of any vector with itself is zero.*

Vector Product/ Cross Product (3)

B. Using Components to Calculate the Vector Product

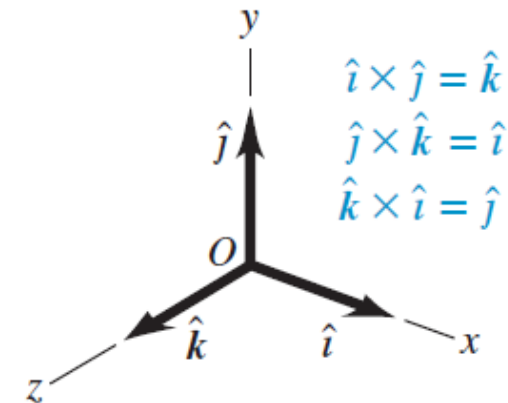
- If we know the components of $\vec{A}$ and $\vec{B}$, we can calculate the components of the vector product by using a procedure similar to that for the scalar product.
- First we work out the multiplication table for unit vectors $\hat{i}$, $\hat{j}$ and $\hat{k}$, all three of which are perpendicular to each other as shown in figure. The vector product of any vector with itself is zero, so

$$\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = \mathbf{0}$$

$$\hat{i} \times \hat{j} = -\hat{j} \times \hat{i} = \hat{k}$$

$$\hat{j} \times \hat{k} = -\hat{k} \times \hat{j} = \hat{i}$$

$$\hat{k} \times \hat{i} = -\hat{i} \times \hat{k} = \hat{j}$$



Vector Product/ Cross Product (4)

- We express $\vec{A}$ and $\vec{B}$ in terms of their components and the corresponding unit vectors, and we expand the expression for the vector product:

$$\begin{aligned}\vec{A} \times \vec{B} &= (A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) \times (B_x \hat{i} + B_y \hat{j} + B_z \hat{k}) \\ &= A_x \hat{i} \times B_x \hat{i} + A_x \hat{i} \times B_y \hat{j} + A_x \hat{i} \times B_z \hat{k} \\ &\quad + A_y \hat{j} \times B_x \hat{i} + A_y \hat{j} \times B_y \hat{j} + A_y \hat{j} \times B_z \hat{k} \\ &\quad + A_z \hat{k} \times B_x \hat{i} + A_z \hat{k} \times B_y \hat{j} + A_z \hat{k} \times B_z \hat{k}\end{aligned}$$

- We can also rewrite the individual terms as mentioned above in previous slide as

$$A_x \hat{i} \times B_y \hat{j} = (A_x B_y) \hat{i} \times \hat{j},$$

- Evaluating these by using the multiplication table for the unit vectors mentioned in previous slide and then grouping the terms, we get:

$$\vec{A} \times \vec{B} = (A_y B_z - A_z B_y) \hat{i} + (A_z B_x - A_x B_z) \hat{j} + (A_x B_y - A_y B_x) \hat{k}$$